

Municipal Data Collection and Compiling

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1 - Background and Methodology

1.1 - Introduction:

The goal for this project was to create a data set focused on municipalities. In this task we worked using public data from Stats Canada due to the inclusion of a DGUID. DGUID, which stands for Dissemination Geography Unique Identifier, is the code varying from 10-21 characters in length used to differentiate each geographic area, introduced in the 2016 Census. Using the DGUID, we can combine the data sets to more easily provide a breadth of data on most municipalities. Within each data set, there are limitations to the number of municipalities each initial data set included, and as such, not all described variables are available for all municipalities.

1.2 - Methodology:

All datasets were sorted and filtered via the use of the Python programming language and the Pandas library, used specifically to collect, sort and transform data in .csv and .xsl files. Programming was used to automatically collect, sort, and filter only the data required from large datasets which contain thousands to millions of rows. It was not feasible to manually collect and process this much data. Examples of some tools used were filtering columns, removing duplicates, and transposing columns and rows.

2 - Data Descriptions:

2.1 - Demographics and Land Area

These datasets were chosen to provide a comparison of each municipality's major features such as land area, dwellings, and population, allowing for more informative and comprehensive data.

2.1.1 - Land Area

This dataset provides the land area of most municipalities in Canada. The main variables that are important to this dataset include Coordinate and Land area in square kilometres, 2021. The Coordinate variable has values that are unique identifiers for a specific area assigned to the table where the dataset is from. The land area column represents the physical size of each municipality, measured in square kilometres. The dataset was last updated in February 2022.

2.1.2 - Population & Dwellings

This dataset presents the changes in population and dwellings for most Canadian municipalities. The main included columns are national population rank in 2021, provincial/territorial population rank in 2021, population change from 2016 to 2021, total dwelling and occupied dwelling change from 2016 to 2021, and also numerical values that can be used to derive the changes between the recorded years, with the last update on the dataset in February 2022.

2.1.3 - Population Data by Age

This dataset contains the population of most municipalities broken down into three age groups in the year 2021. The relevant columns in this dataset include “0 to 14 years”, “15 to 64 years”, and “65 years and older”. The columns contain the population that falls into those three age groups. The dataset was last updated in March 2024.

2.2 - Cultural Identity

These datasets enable a comparison of cultural characteristics across municipalities.

2.2.1 - Immigration

This dataset provides information on immigration status and period of immigration, including non-immigrants and non-permanent residents across Canada in the year 2021. The main columns include Non-Immigrants, Immigrants, Recent Immigrants (2016-2021), and Non-Permanent Residents. Non-Immigrants count individuals born in Canada that do not have immigrant status. Immigrants refers to people who have moved to Canada, while Recent Immigrants (2016-2021) captures those who arrived within that time frame. Non-Permanent Residents include individuals residing temporarily in Canada. The dataset was last updated in March 2026.

2.2.2 - Visible Minorities

This dataset outlines the population size for each minority group as well as the share they represent within each municipality in the year 2021. The relevant columns in this dataset include the Total Minority Population, main visible minority groups, Minority, n.i.e, and Multiple Minorities. Total minority population represents the total number of individuals identified as visible minorities. Main visible minority groups show the percentage share of each major visible minority group within the total population, and are represented by each distinctive column. Minority, n.i.e captures the percentage of individuals that identify as a visible minority but are not included in the main categories. Multiple Minorities refers to the percentage of individuals that are a part of more than one visible minority group. The dataset was last updated in June 2023.

2.2.3 - Language

This dataset contains the languages spoken by the population of each municipality in 2021. The populations are broken down into columns by English only, French only, both, or neither. The total population across all language categories is also provided. The dataset was last updated in March 2026.

2.3 - Socio-Economic

This dataset provides a comparison of each municipality's economic features.

2.3.1 - Labour Force Characteristics

This dataset shows the average unemployment rate for each locality along with its corresponding province in 2021. The key columns in this dataset are Province and Average Unemployment Rate (2021). Province indicates the province in which each locality is located, while Average Unemployment Rate (2021) shows the average percentage of the labour force that was unemployed in 2021. The dataset was last updated in June 2024.

3 - Conclusion:

Overall, data was filtered and processed with the aid of Python and the Pandas library to simplify the overall process and create a more efficient workflow. All datasets were collected from the Statcan website before being processed to extract the required information. The final compiled data includes land area, population and dwelling counts, population data by age, immigration status, visible minorities, and language. All data collected is for the census year 2021, and is collected from a reputable source, the government of Canada.